

Mini Assignment: Dictionary in Python

1. Student Performance Analytics System

Problem Statement

A coaching institute wants to analyze student performance.

Store details of at least **30 students** in a dictionary.

Example Structure

```
students = {  
    "S101": {"name": "Anuj", "marks": 85},  
    "S102": {"name": "Rahul", "marks": 72}  
}
```

Requirements

1. Display all student records.
2. Search a student using Student ID.
3. Add a new student.
4. Update marks of an existing student.
5. Delete a student.
6. Find topper and lowest scorer.
7. Calculate class average.
8. Count pass and fail students.
9. Generate grades:
 - A (90+)
 - B (75–89)
 - C (50–74)
 - F (<50)
10. Display students scoring above average.
11. Display top 5 performers.
12. Create a separate dictionary for scholarship students (marks > 85).

Expected Learning

- Nested Dictionaries
- Dictionary Traversal
- Searching
- Aggregation
- Report Generation

2. E-Commerce Inventory & Sales Dashboard

Problem Statement

An online store wants to manage products and sales.

Example Structure

```
products = {  
    "P101": {  
        "name": "Laptop",  
        "price": 55000,  
        "stock": 12,  
        "sold": 25  
    }  
}
```

Maintain records of at least **30 products**.

Requirements

1. Display all products.
2. Add a new product.
3. Update stock after sales.
4. Find out-of-stock products.
5. Find products with stock less than 5.
6. Calculate total inventory value.
7. Find best-selling product.
8. Find least-selling product.
9. Calculate total revenue generated.
10. Generate a low-stock report.
11. Display products whose sales exceed the average sales.
12. Create a dictionary of products eligible for promotion (sales < 10).

Challenge

Generate a complete business report.

3. Smart Library Management System

Problem Statement

Create a digital library management system.

Example Structure

```
library = {  
    "B101": {  
        "title": "Python Basics",  
        "author": "ABC",  
        "copies": 5  
    }  
}
```

Maintain records of at least **30 books**.

Requirements

1. Add a book.
2. Remove a book.
3. Search a book by ID.
4. Search by title.
5. Update available copies.
6. Issue a book.
7. Return a book.
8. Display books with fewer than 3 copies.
9. Display books that are unavailable.
10. Find the most available book.
11. Generate a restocking report.
12. Create a separate dictionary of books requiring immediate purchase.

Challenge

Generate a complete library summary report.

4. Cricket Tournament Analytics System

Problem Statement

Store statistics of at least **30 cricket players**.

Example Structure

```
players = {  
  "Virat": {  
    "runs": 645,  
    "matches": 12,  
    "wickets": 0  
  }  
}
```

Requirements

1. Display all player statistics.
2. Find highest run scorer.
3. Find lowest run scorer.
4. Calculate average runs.
5. Find player with maximum wickets.
6. Find all-rounders (runs > 300 and wickets > 5).
7. Display players scoring above average.

8. Create categories:

- Star Performer
- Good Performer
- Average Performer
- Poor Performer

9. Generate team statistics.

10. Display top 5 batsmen.

11. Display top 5 bowlers.

12. Create a separate dictionary for award winners.

Challenge

Generate a tournament report.

5. City Population & Development Dashboard

Problem Statement

The government wants to analyze city data.

Store details of at least **30 cities**.

Example Structure

```
cities = {  
    "Delhi": {  
        "population": 32000000,  
        "area": 1484,  
        "literacy": 89  
    }  
}
```

Requirements

1. Display all city details.
2. Find the most populated city.
3. Find the least populated city.
4. Calculate average population.
5. Display cities with literacy rate above 90%.
6. Display cities with literacy below average.
7. Calculate population density.
8. Find city with highest density.
9. Categorize cities:
 - Small
 - Medium

- Large

10. Create a development-priority list.

11. Generate separate dictionaries for:

- High Literacy Cities
- Low Literacy Cities

12. Generate a national summary report.

Challenge

Rank all cities based on population density.

Assignment Rule (Important)

For all 5 Questions:

- Use **at least 30 records**.
- Do **not use built-in sorting functions** (`sorted()`, `sort()`).
- Use loops and conditions to find maximum, minimum, rankings, and reports.
- Display results in a structured format.
- Add a menu-driven interface using while loop.